

WEEK 5 : ADVANCED MACHINE LEARNING

Performance Comparison Report

1. Problem Statement

The goal of this project is to predict residential house prices based on structural features and locational attribute of the property (proximity to roads, number of bedrooms, availability of parking area .e.t.c)

2.Dataset Description

The Dataset (“housing.csv”) contains 545 records with 13 columns: Price(target variable), area, bedrooms, bathrooms, stories, parking, airconditioning, prefarea, mainroad, guestrooms, basement, hotwaterheating, furnishingstatus. No missing values or duplicate rows were present.

Pre-processing steps carried out before modelling:

- Binary columns were encoded as 0/1
- Area and Price were log-transformed to reduce skewness
- Furnishing Status was encoded ordinally
- Features were standardized using StandardScaler for Linear Model, tree-based models were trained on unscaled features
- random_state = 42 for consistent reproducible comparison across all models

3. Models Used

- Baseline: Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor

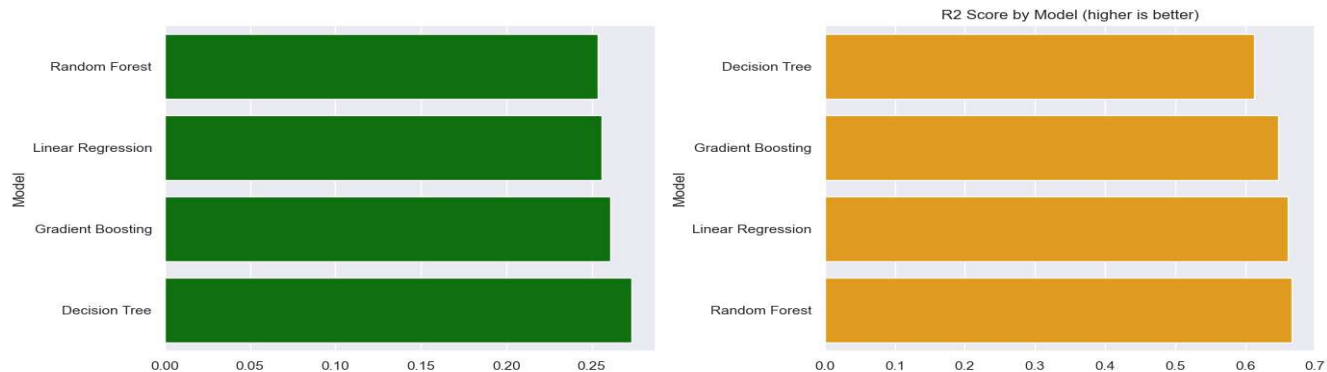
4. Evaluation Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error(MSE)
- Root Mean Squared Error(RSME)
- R2 Score(R2)

5.Performance Comparison

MODEL	MAE	MSE	RSME	R2
Linear Regression	0.2061	0.0655	0.2560	0.6605

Decision Tree	0.2125	0.0748	0.2736	0.6124
Random Forest	0.1970	0.0644	0.2536	0.6667
Gradient Boosting	0.2026	0.0680	0.2609	0.6476
Gradient Boosting (Tuned)	0.1978	0.0645	0.2539	0.6661



6. KEY FINDINGS

- Random Forest with default Parameters was the best performing Model($R^2 = 0.6667$)
- The Single Decision Tree was the weakest Model($R^2 = 0.6124$)
- Hyperparameter Settings did not beat the default Models